

Lab H: Third Arduino Lab! Writing Strings to the OLED.

Due: March 17, 9 p.m.

1 Introduction

In the last lab we handled events that we detected on our Arduino board. In this lab, we will again detect and respond to events. We will also get some more practice with String formatting, and we will write some Strings to the OLED output on our boards.

The OLED display on the Grove Beginner Kit is one of a family of similar displays found in many products. Learning to use this display will permit you to use this or similar displays in your engineering projects.

There are three relevant main sites for data on the OLED display found on the Grove Beginner Kit for Arduino (SSD1315):

1. Seeed Studio's SSD1515 page
2. The Firmata4j SSD1306 class
3. Adafruit's page on these types of displays

2 Programming Task

To start, you will be altering the file called *SecondButtonListener.java* and completing both a constructor and methods to handle some string processing.

More specifically, in the *SecondButtonListener* class, you will implement:

1. The method *cleanSentence(String input)*. This method should accept a String as input and return a String that is the same as the input yet without any duplicate words and with words that are separated by a comma and a space. For example if the input is "Java is fun and Java is powerful" the output should read "Java, is, fun, and, powerful". You may want to consider using a Set ADT in order to remove duplicate words from your string; you can read more about Sets at this URL.
2. The method *hashSentence(String input)*. This method should accept a String as input and return a *HashMap<String, Integer>* object that maps each word in the sentence onto the number of times it appears in the sentence. To learn more about the HashMap ADT in Java visit this URL.

3. The method *formatList(ArrayList<String> input)*. This method should accept an ArrayList of Strings as input and return an *String* object that formats every entry in the input ArrayList into a numbered list. For example if the input ArrayList contains the strings “Shahin”, “Sonya” and “Mohammed”, the output String should read:

1. Shahin
2. Sonya
3. Mohammed

To learn more about the ArrayList ADT in Java visit this URL.

4. Finally, implement the overridden method *onPinChange(IOEventevent)*. This is an example of an event handler; it will be executed when a Pin that is listening for changes detects a change. Within this method, check for the value of the *button* attribute. If it is 1, call your *formatList* method on an array containing the words: “Civil”, “Electrical”, “Mechanical” and “Software”. Then, print the output of this method to the OLED display on the Grove board. If the value of *button* is 0, clear the OLED display.

Then, in the *SecondButtonApp* class:

- Complete the constructor *SecondButtonApp()*. More specifically, you will need to initialize a Pin object with the value of pin 6 (the button) and an on your Grove Kit. Set the mode of the button pin to Pin.Mode.INPUT. Finally, create a *SecondButtonListener* and add it an event listener on the grove board using the method *addEventListener*. Use the Pin you initialized as well as the reference to the OLED object as input arguments to the *SecondButtonListener* constructor.

3 Testing your Classes

As always, a small number of test cases have been provided to help you test your implementations. Make sure you pass these tests! And, as ever, make sure you write your own tests, too.

4 What to Submit

1. SecondButtonListener.java
2. SecondButtonApp.java

HAVE FUN AND GOOD LUCK!